

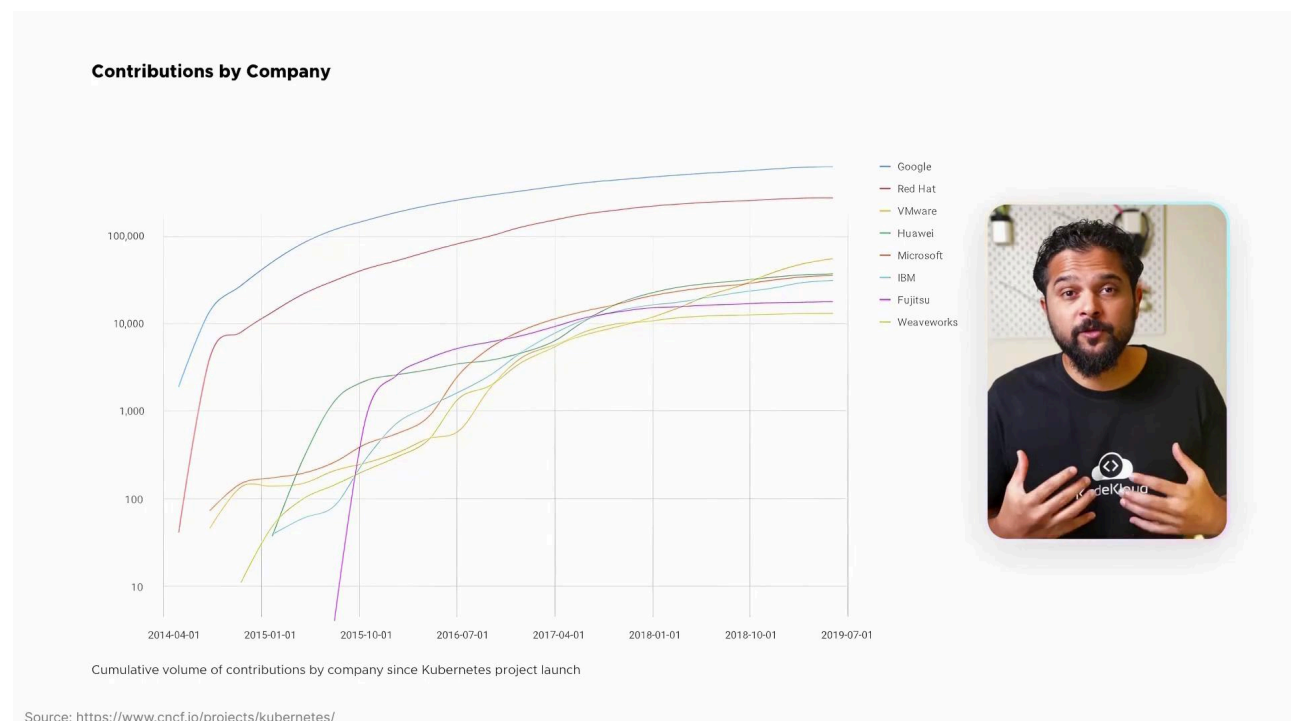
Kubernetes and Cloud Native Associate - KCNA

Cloud Native Architecture

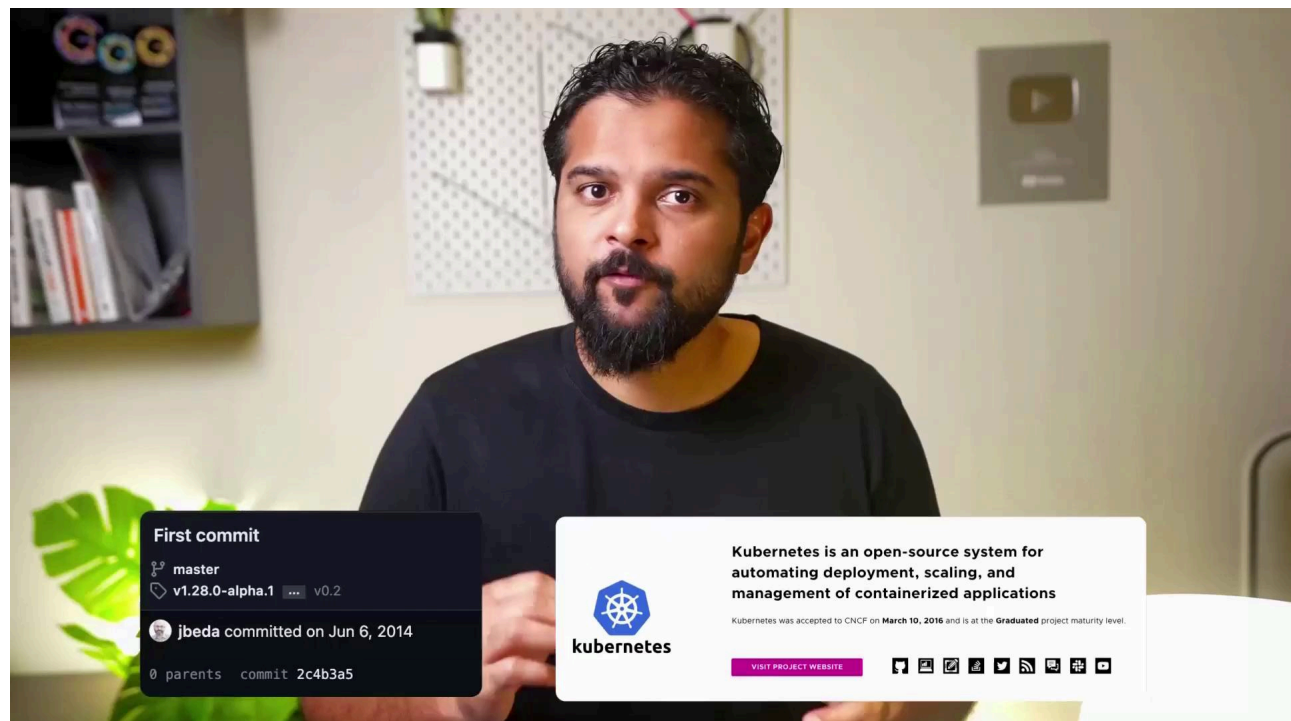
Kubernetes SIG

Kubernetes is one of today's most popular open source projects, boasting over 80,000 stars on GitHub, contributed to by more than 2,500 developers and over 1 million contributions. This powerful container orchestration platform is the backbone for managing modern, container-based applications.

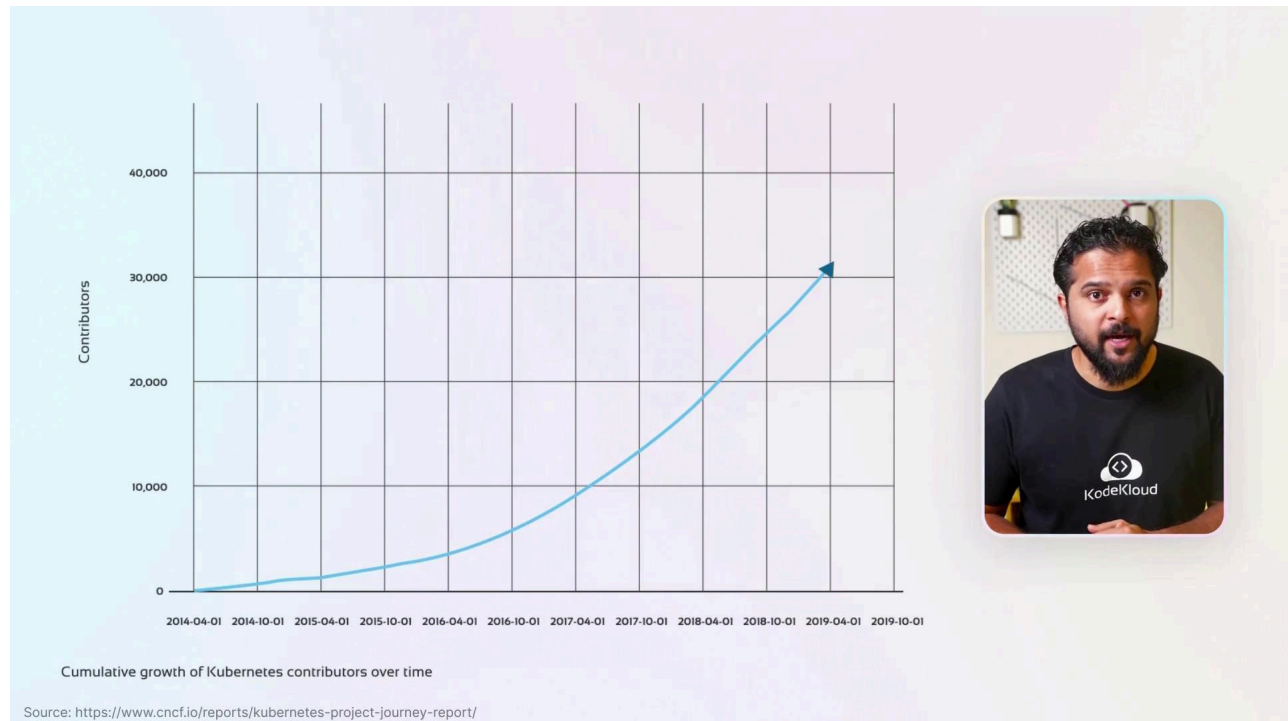
Big industry players such as Google, Microsoft, and AWS rely on Kubernetes not only for their container management needs but also for its innovative hosting solutions and significant contributions to its development.



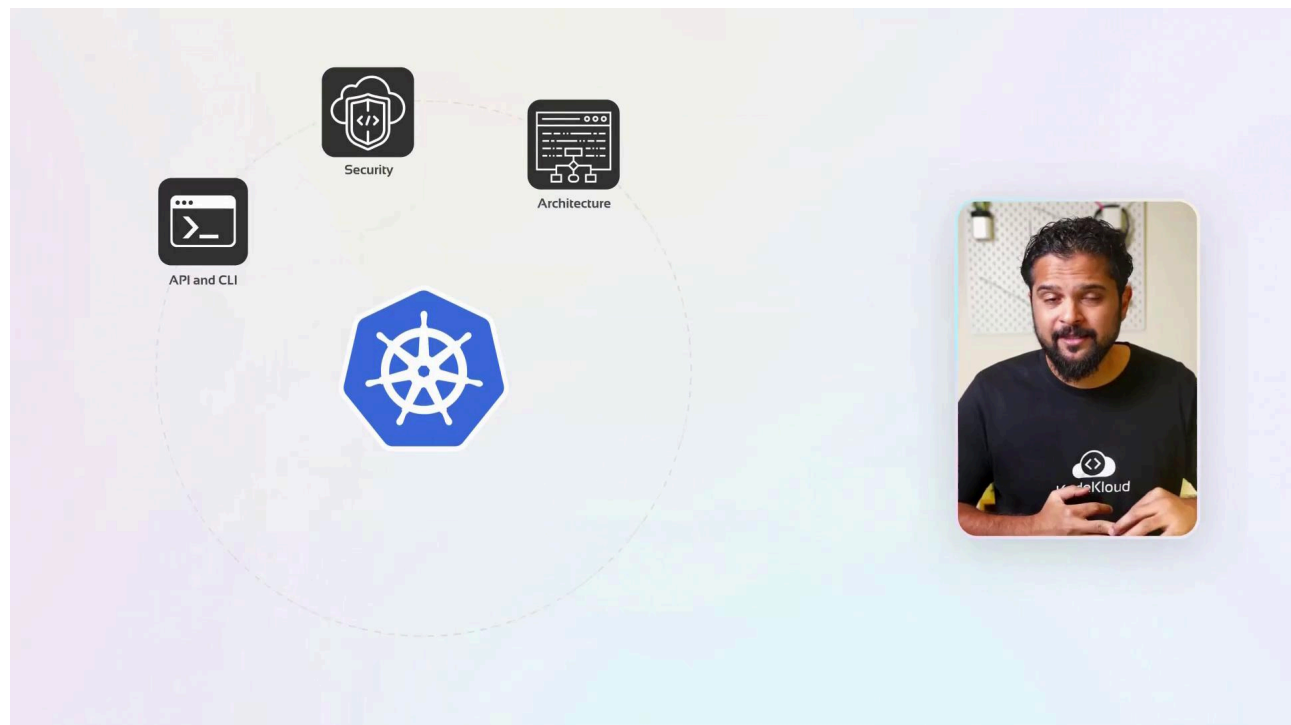
Kubernetes was originally developed by Google in 2014 to simplify the deployment and management of containerized applications across multiple hosts. The project's first commit was on June 6th, 2014, and its evolution accelerated when it joined the Cloud Native Computing Foundation (CNCF) in 2016, reinforcing its importance in container orchestration.



Starting with just a few developers, Kubernetes has experienced exponential growth. Initially, there were around 20 developers; this number increased to approximately 400 after joining the CNCF. Today, over 3,000 contributors drive the project forward. The following chart from a CNCF report illustrates the cumulative growth of Kubernetes contributors over the years.



Managing a project as massive as Kubernetes is complex. Imagine constructing a skyscraper in a busy city: coordinating construction workers, managing supply chains, and ensuring safety all at once. Similarly, Kubernetes requires meticulous handling of architecture, security, APIs, command line interfaces (CLI), autoscaling features, networking, and storage solutions—especially with integrations across various cloud providers.

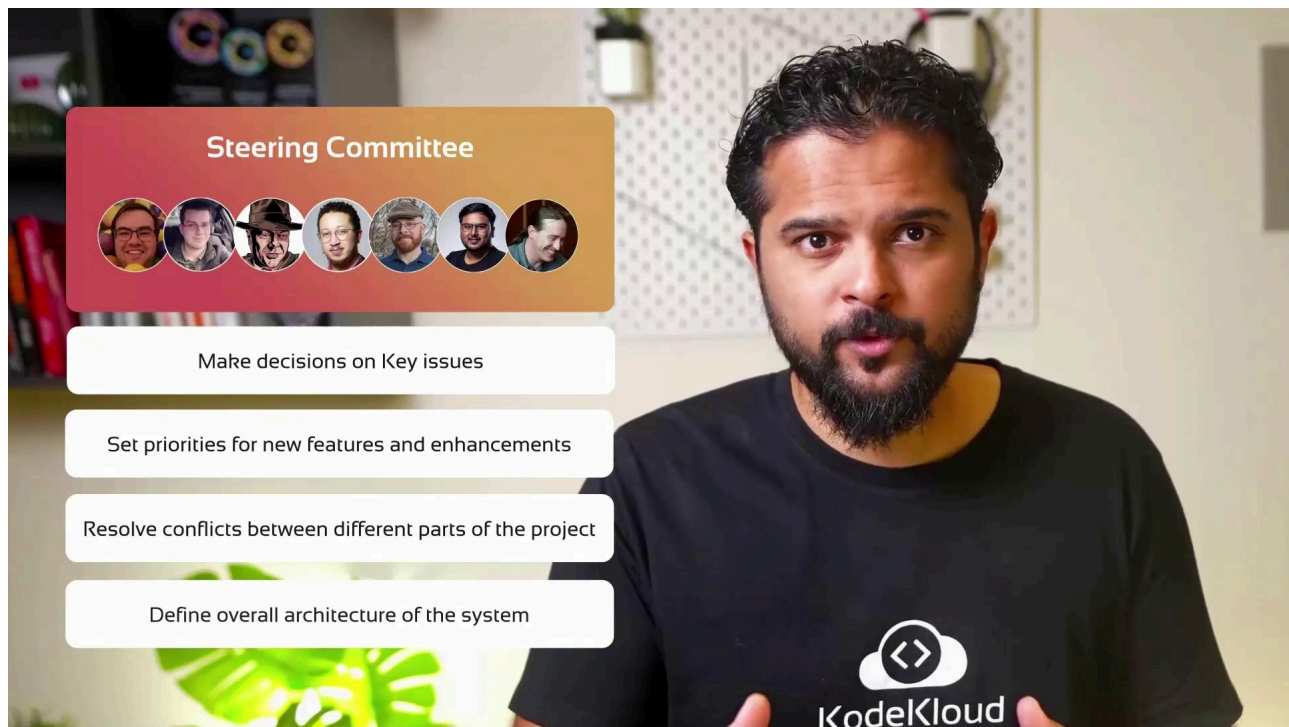


Project Management Insight

A robust project management system is essential for coordinating feature development, bug fixes, testing, and release cycles within the Kubernetes ecosystem.

Due to its open source nature, Kubernetes is managed by a diverse community rather than a centralized organization. The success of this approach lies in its well-defined community governance model.

At the top of this hierarchy is the Kubernetes Steering Committee. This diverse group is responsible for setting the overall project direction, defining system architecture, prioritizing new features, and resolving conflicts across different areas. They provide guidance to various working groups and Special Interest Groups (SIGs), which are integral to Kubernetes' operational structure.



As of this recording, the Steering Committee consists of:

- Benjamin Elder (Google)
- Christoph Blecker (Red Hat)
- Carlos Taddeu Panato Jr. (ChainGuard Inc.)
- Stephen Augustus (Cisco)
- Bob Killen (Google)
- Nabarun Pal (VMware)
- Tim Pepper (VMware)

Following the Steering Committee are various working groups and SIGs. While working groups address cross-cutting challenges spanning multiple domains, SIGs serve as specialized teams responsible for distinct facets of Kubernetes.

Consider the Kubernetes SIG for Architecture—it functions similarly to a team of architects designing a skyscraper, focusing on evolving design principles and ensuring a consistent architectural approach throughout the platform. SIGs streamline development, foster rapid innovation, and maintain organizational clarity by avoiding overlap between responsibilities.

Key responsibilities of SIGs include:

1. **Code Development:** Introducing new features, resolving bugs, and improving the codebase.
2. **Testing and Validation:** Ensuring that releases meet rigorous community quality standards.
3. **Documentation:** Keeping user guides, reference materials, and API documentation up to date.
4. **Community Outreach and Education:** Organizing meetups, webinars, and conferences to engage and educate users.
5. **Release Management:** Coordinating the entire release process, from feature development to documentation updates.
6. **Architecture and Design Guidance:** Ensuring Kubernetes remains scalable, reliable, and maintainable.

SIGs conduct their discussions openly via video conferences, chat rooms, mailing lists, Slack groups, GitHub issues, and pull requests. All meeting details are accessible on the public Kubernetes community calendar, with recordings available online for later review.

Technical proposals and design changes are managed through the Kubernetes Enhancement Proposal (KEP) process. KEPs invite community feedback as SIG members review proposals through GitHub, mailing lists, and dedicated SIG meetings.

Each SIG is led by one or more chairs who facilitate discussions and drive decision-making. A comprehensive list of SIGs, including their co-chairs, communication channels, and meeting schedules, is publicly available.

OWNERS_ALIASES

README.md

SECURITY.md

SECURITY_CONTACTS

SIG-diagram.png

SIG-diagram.svg

code-of-conduct.md

community-membership.md

go.mod

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governance.md

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liaisons.md

sig-list.md

sig-wg-lifecycle.md

sig.yaml

values.md

Documentation • Share feedback

Community Groups

Community Groups

Interest Groups, Working Groups, User Groups and Committees.

99

docs

/groups

/sigs

community-groups

Most community activity is organized into Special Interest Groups (SIGs), time bounded Working Groups, and the community meeting.

SIGs follow these [guidelines](#) although each of these groups may operate a little differently depending on their needs and workflow.

Each group's material is in its subdirectory in this project.

When the need arises, a new SIG can be created

Special Interest Groups

Name	Label	Chairs	Contact	Meetings
API Machinery	api-machinery	- David Eads, Red Hat - Federico Bongiovanni, Google	- Slack - Mailing List	- Kubebuilder and Controller Runtime Meeting: Thursdays at 11:00 PT (Pacific Time) (biweekly) - Regular SIG Meeting: Wednesdays at 11:00 PT (Pacific Time) (biweekly)
Apps	apps	- Janet Kuo, Google - Kenneth Owens, Snowflake - Maciej Szulik, Red Hat	- Slack - Mailing List	Regular SIG Meeting: Mondays at 9:00 PT (Pacific Time) (biweekly)
Architecture	architecture	- Derek Carr, Red Hat - Davanum Srinivas, Amazon Web Services - John Belamaric, Google	- Slack - Mailing List	- Enhancements Subproject Meeting: Thursdays at 10:00 PT (Pacific Time) (biweekly) - Production Readiness Office Hours: Wednesdays at 12:00 PT (Pacific Time) (biweekly) - Regular SIG Meeting: Thursdays at 11:00 PT (Pacific Time) (biweekly) - code organization Office Hours: Thursdays at 14:00 PT (Pacific Time) (biweekly) - conformance office Hours: Wednesdays at 18:00 UTC (First Wednesday of the month)
Auth	auth	- Mo Khan, Microsoft - Mike Danese, Google - Rita Zhang, Microsoft	- Slack - Mailing List	- Regular SIG Meeting: Wednesdays at 11:00 PT (Pacific Time) (biweekly) - Secrets Store CSI Meeting: Thursdays at 8:00 PT (Pacific Time) (biweekly) - Weekly Issues/PR Triage Meeting: Mondays at 9:00 PT (Pacific Time) (weekly)
Autoregulation	autoregulation	Guy Templeton, Skyscanner	Slack	Regular SIG Meeting: Mondays at 10:00 PT (Pacific Time) (weekly)

For example:

- SIG Architecture:** Oversees Kubernetes' overall design and API consistency. At the time of recording, it is chaired by Derek Carr (Red Hat), Davanum Srinivas or DIMMS (AWS), and John Bellamarak (Google).
- SIG Cluster Lifecycle:** Manages the creation, management, and upgrading of clusters. It is chaired by Justin Santabarbero (Google) and Vince Brignano (Red Hat).
- SIG Storage:** Focuses on storage management, ensuring consistent API definitions across storage providers, chaired by Saad Ali (Google) and Jing Yang (VMware).
- SIG Network:** Handles networking functionalities with a consistent API across providers, chaired by Michael Zappa (Microsoft), Shane (Kong), and Tim Hockin (Google).

New SIGs or membership expansion begins with community proposals, which are reviewed by the Kubernetes Steering Committee to ensure alignment with overall community goals. Once approved, the SIG adopts its own governance structure through a nomination and election process, electing its leaders accordingly.

This collaborative model has enabled Kubernetes—a global open source project—to flourish through collective contributions.

The screenshot shows a code editor displaying the file `community / sig-list.md`. The file content includes a table of Special Interest Groups (SIGs) and a video call overlay featuring a man with a beard wearing a KodeKloud t-shirt.

title	linkTitle	description	weight	type	aliases	slug
Community Groups	Community Groups	A list of our community groups: Special Interest Groups, Working Groups, User Groups and Committees.	99	docs	/groups /sigs	community-

Most community activity is organized into Special Interest Groups (SIGs), time bounded Working Groups, and the code of conduct.

SIGs follow these [guidelines](#) although each of these groups may operate a little differently depending on their needs.

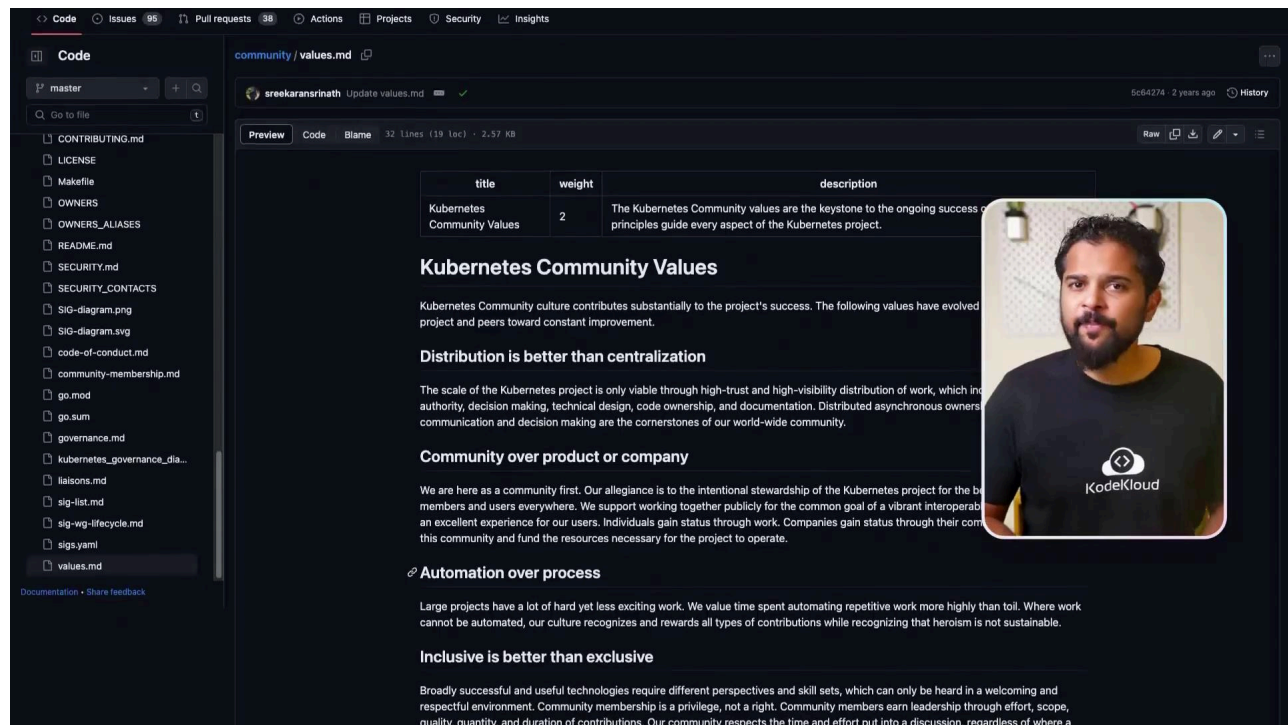
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In summary, the Kubernetes community thrives on open collaboration, technical excellence, and inclusivity. Whether your interests lie in code development, testing, documentation, or community outreach, there is a SIG that aligns with your expertise. Participation is highly encouraged—join the Kubernetes Slack channel and mailing lists, attend SIG meetings, and contribute to one of the most dynamic open source projects available.



The screenshot shows a GitHub repository for the Kubernetes SIG. The file being viewed is `community/values.md`, which was updated by `sreekaransrineth` 2 years ago. The file contains a table of values and several paragraphs of text.

title	weight	description
Kubernetes Community Values	2	The Kubernetes Community values are the keystone to the ongoing success of the project. These principles guide every aspect of the Kubernetes project.

Kubernetes Community Values

Kubernetes Community culture contributes substantially to the project's success. The following values have evolved over time and peers toward constant improvement.

Distribution is better than centralization

The scale of the Kubernetes project is only viable through high-trust and high-visibility distribution of work, which includes authority, decision making, technical design, code ownership, and documentation. Distributed asynchronous ownership, communication and decision making are the cornerstones of our world-wide community.

Community over product or company

We are here as a community first. Our allegiance is to the intentional stewardship of the Kubernetes project for the benefit of all members and users everywhere. We support working together publicly for the common goal of a vibrant interoperable ecosystem, an excellent experience for our users. Individuals gain status through work. Companies gain status through their contributions to this community and fund the resources necessary for the project to operate.

Automation over process

Large projects have a lot of hard yet less exciting work. We value time spent automating repetitive work more highly than toil. Where work cannot be automated, our culture recognizes and rewards all types of contributions while recognizing that heroism is not sustainable.

Inclusive is better than exclusive

Broadly successful and useful technologies require different perspectives and skill sets, which can only be heard in a welcoming and respectful environment. Community membership is a privilege, not a right. Community members earn leadership through effort, scope, quality, quantity, and duration of contributions. Our community respects the time and effort put into a discussion, regardless of where a

We hope this article has provided clear insights into Kubernetes SIGs and their role in driving innovation while upholding robust quality standards. Explore further by joining your preferred SIG, participating in community discussions, and contributing to one of the most influential open source projects in existence.



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